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Software Testing & Quality Assurance

Q. Answer the following questions -

1) Explain Seven Testing principles in details.
= Software Testing principles -
I) Testing shows presence of defects -
The goal of Software testing is to make the Software fail. Software testing is to reduces the presence of defects. Software testing takes about the presence of defects & doesn't talk about the presence of defects.
2) Exhaustive Testing is not possible -
It's the process of testing the functionality of the Software in all possible inputs & pre-conditions is known as exhausting testing.
Exhaustive testing is impossible means the Software can never test at every test case. It can test that the Software is correct and it will produce the correct o/p in every test case.
3) Early testing -
To find the defect in the Software early test activity shall be started.
The defect detected in early phases of SDLC will be very less expensive.
for better performance of Software, Software testing will start at the initial phase i.e. testing will perform at the requirement analysis phase.
4) Defect Clustering -
In a project, a Small number of module can contain most of the defects.

The pareto principles for Software testing States that 80 % of Software defects come from 80 % of modules.

5) Pesticide Paradox –

Repeating the same test cases again and again, will not find new bugs. So it is necessary to review the test cases & add or update test case to find new bugs.

6) Testing is context dependent –

The testing approach depends on the context of the software developed. Different types of Software need to perform on different types of testing.

7) Absence of Errors fallacy –

If a built Software is 99 % bug free but does not follow the user & requirement then it is Unusable. It is ~~but~~ not only necessary that Software is 99 % bug-free but it is also mandatory to full fill all the customer requirements.

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2) What is V-model and W-Model? Explain?
= V- Model-

The V- Model is a graphical representation of a Systems development lifecycle.
It is used to preduce regorous development life cycle models.
A contempory of traditional software development is V- model V-model also reffered to as the verification and validation modal.
The curx of V-modal establishes on association between each phase of testing with that of developer.
The phases of testing are categories as Validation phase & that of development
The V- Shaped modal should be used when the requirement is well defined and not ambigous.
This Saves a lot of time.

W- modal-
This modal is introduced by Paul Herzlich. It indicates the one-to-one relationship that exists between the document and test activities.
W- model is a sequential apperance to test a product. It can be done only once the development of the product is completed with no modification required.

W-model is a variant of the V-model.
It is ~~Specia~~ specificaly ~~focud~~ focused on identifying product risks concentrate on points where testing can be most effective.
The testing process is carried out parallel to development process.
It is not first started & after the development is complete.

3) Define levels of Testing.
— There are mainly four levels of Testing —

I) Unit Testing —
The primary goal of testing is to take the smallest piece of testable software in application to isolate it from remainder of code & determine weather it behaves exactly as you expect.

II) Integration Testing —
The integration testing emphases on finding defects which memory mainly arise because of combining various components for testing.
The main objectives of integration testing is to take unit test components & build a program structure
It has 3 types — 1) Big-Bang 2) Top-down and 3) Bottom-up.

III) System Testing –

Once the entire System has been built then it has to be tested against the System specification to check if it delivers features required.
System testing verifies the entire product after integrating all s/w and h/w components and validates it according to original project requirement.

IV) Acceptance Testing –

Acceptance testing is often final step before calling out application. It is performed to determine whether system meets user requirements or not.
Acceptance tests can range from an informal "test drive" to a panned & systematically executed series of tests.